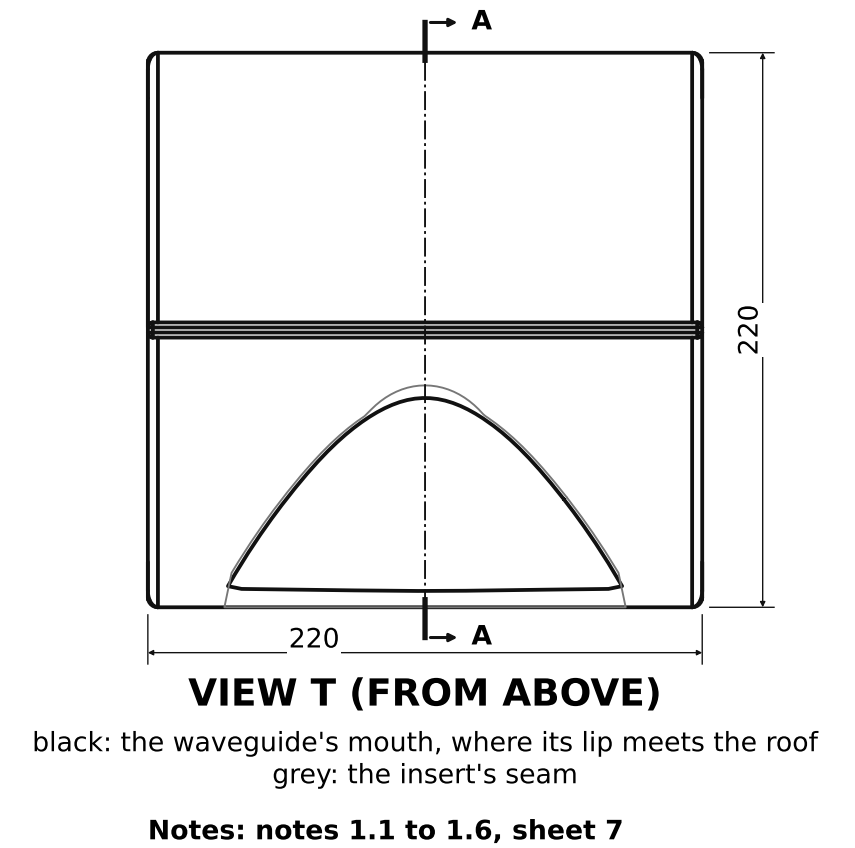
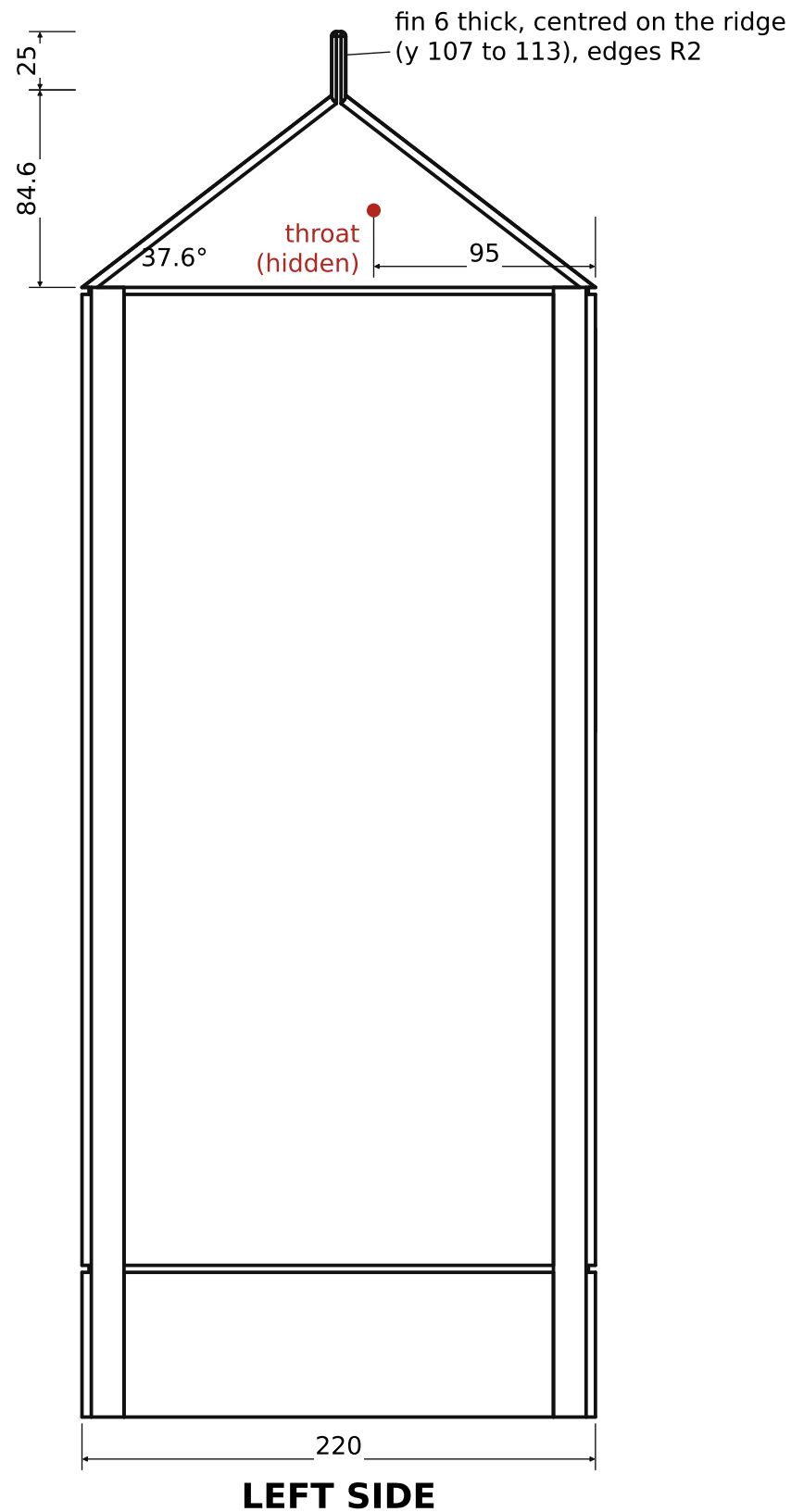
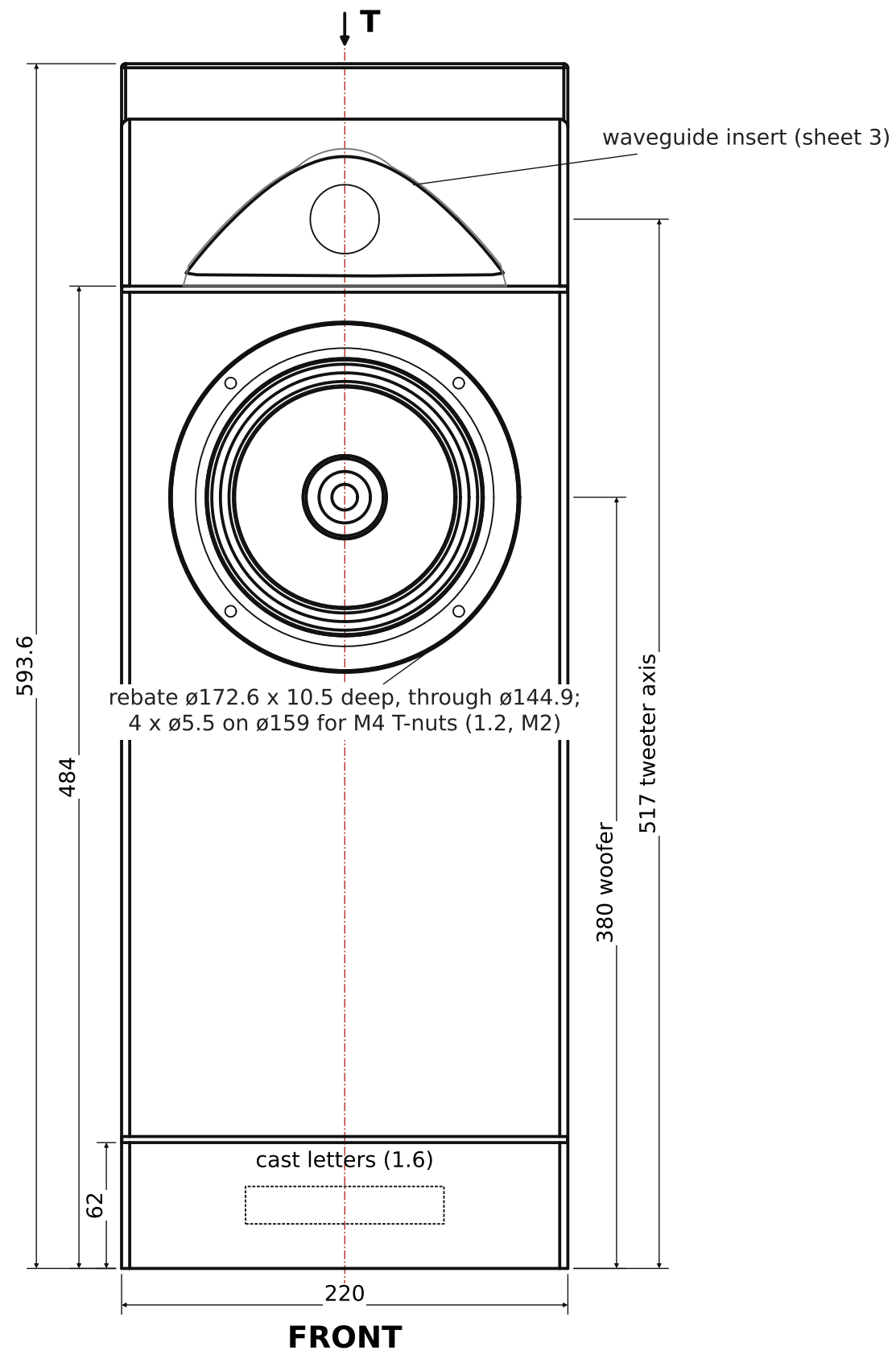


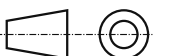


General arrangement

earmilk bookshelf. Front, left side and top at 1:3. Heights from the floor; depths from the front face.

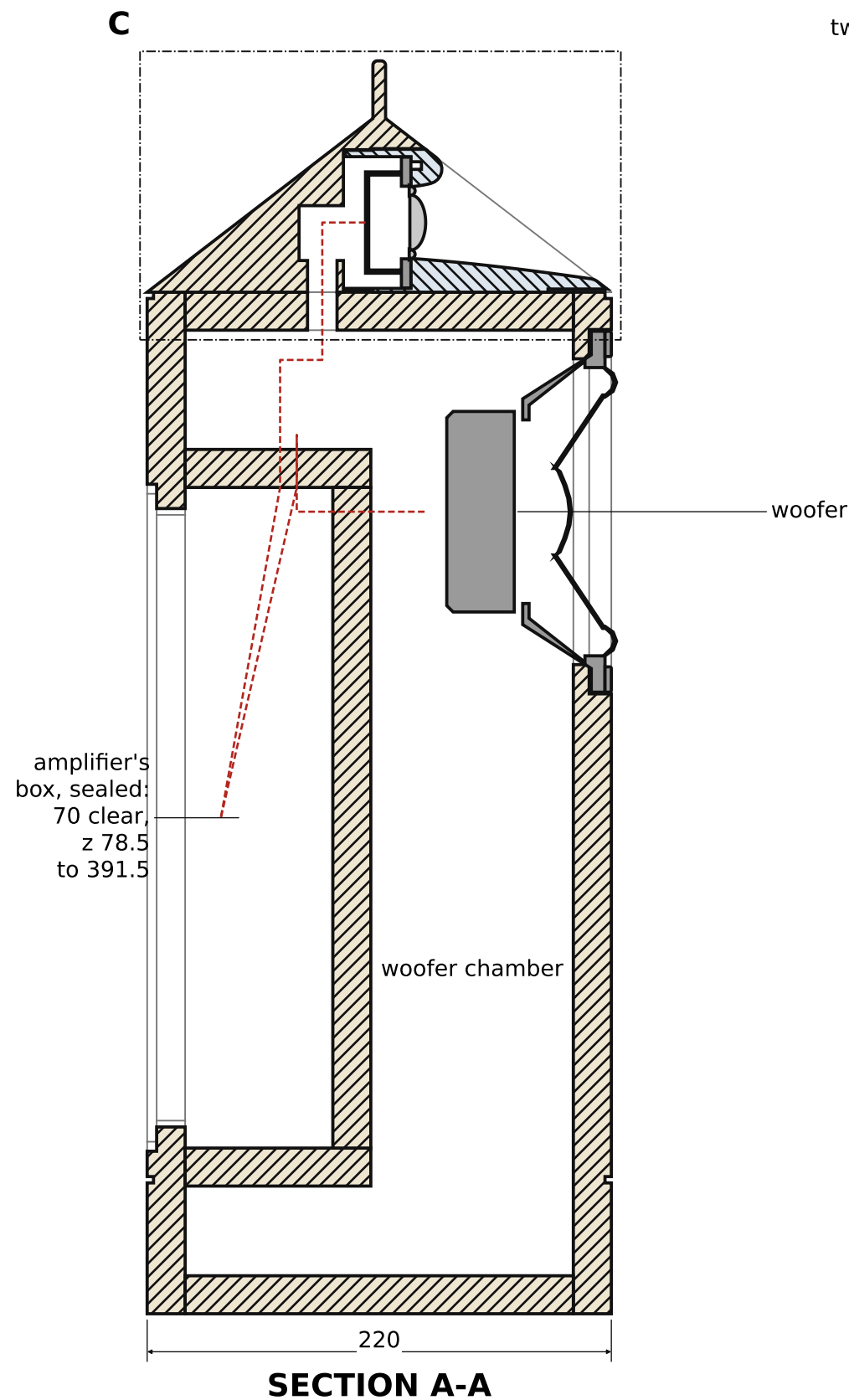
mm | issue 2026-10-09 | from fab/sheets.py (the CAD)

Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

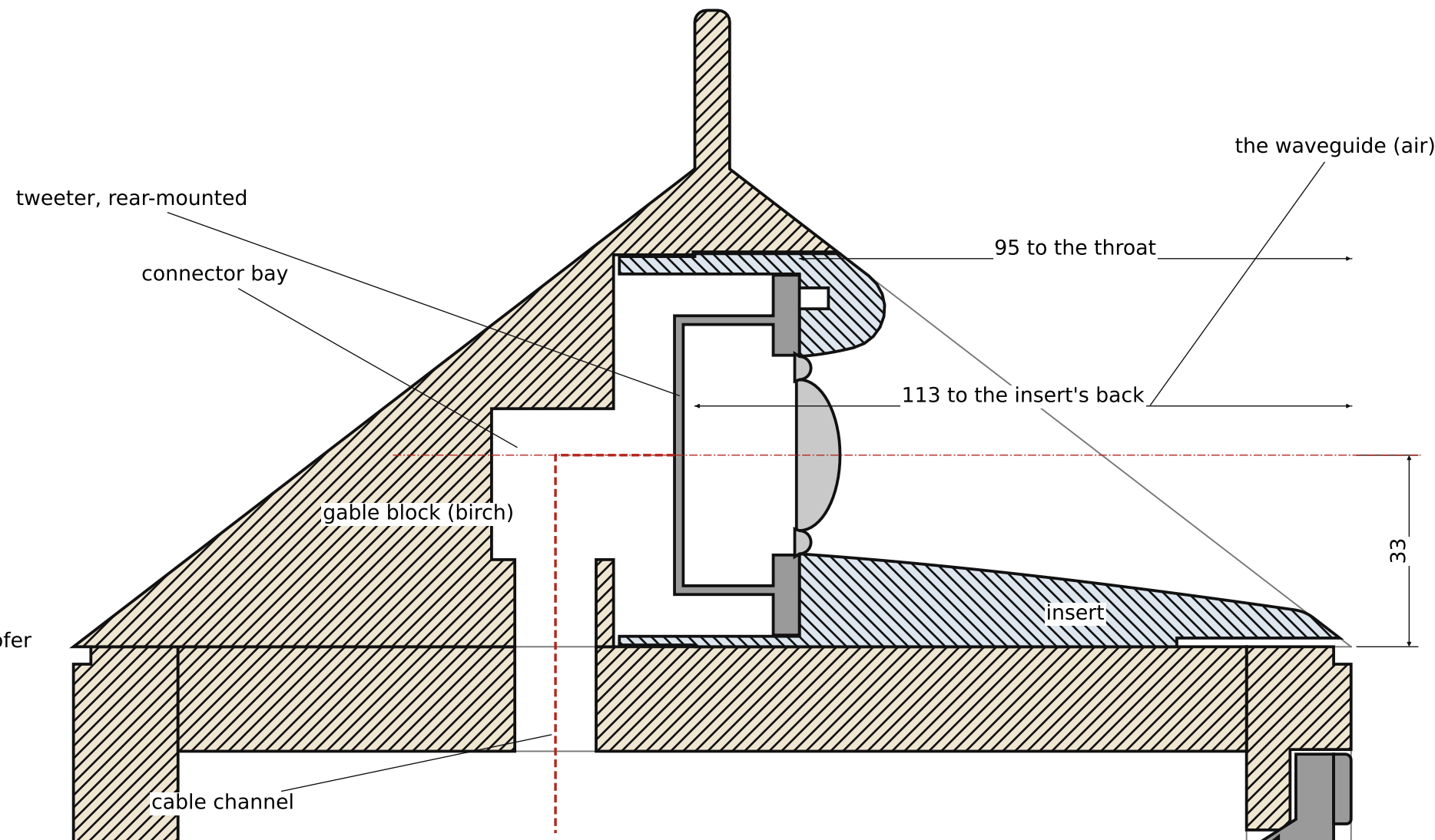


1 of 8

EM-BS-001 rev D



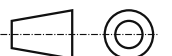
red dashed: the cables' runs to the amplifier (sheet 4)



The roof: notes 2.1 to 2.4, sheet 7

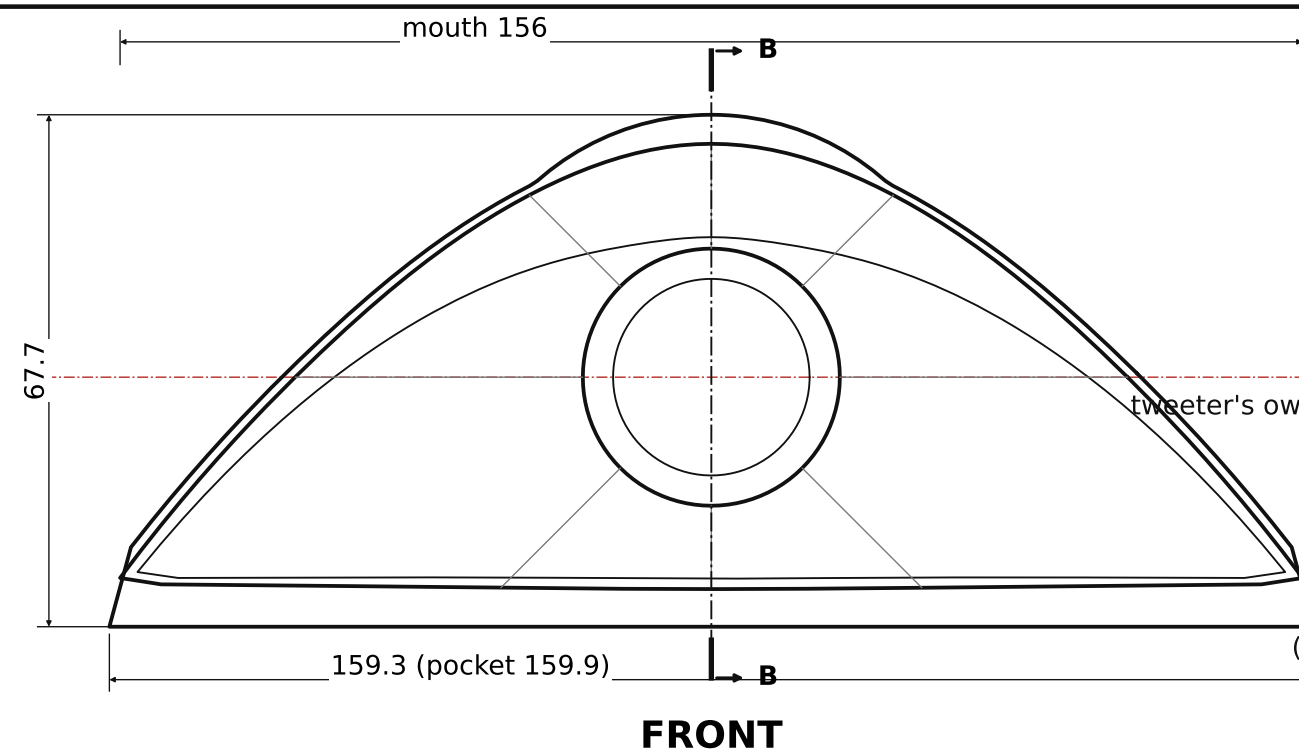
Section on the centreline

earmilk bookshelf. Section A-A on the centre plane (x 110), from the left, at 1:3; the roof at 1:1. The insert hatched the other way.

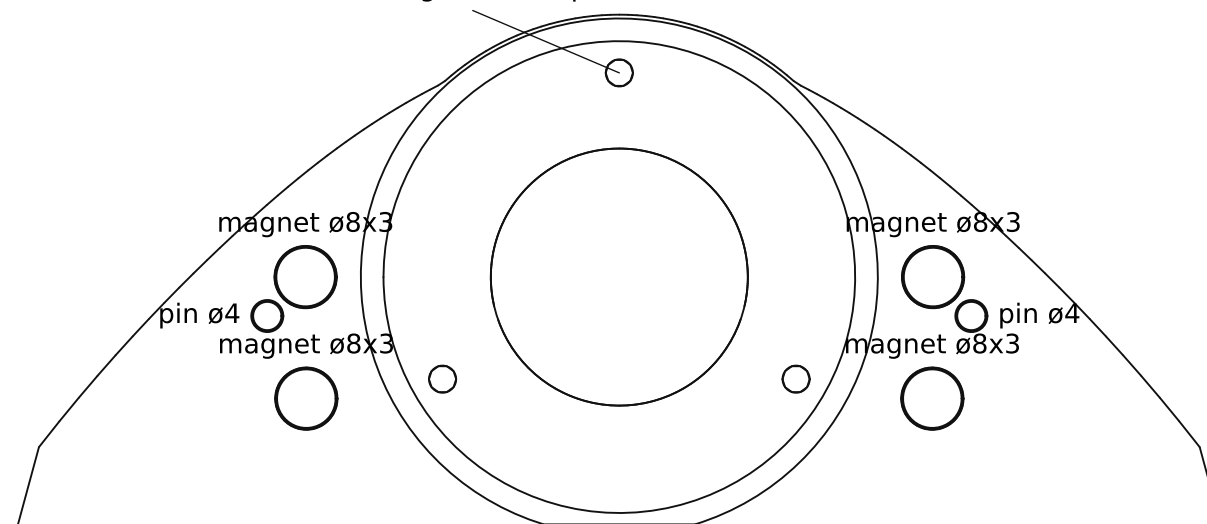


mm | issue 2026-10-09 | from fab/sheets.py (the CAD)
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

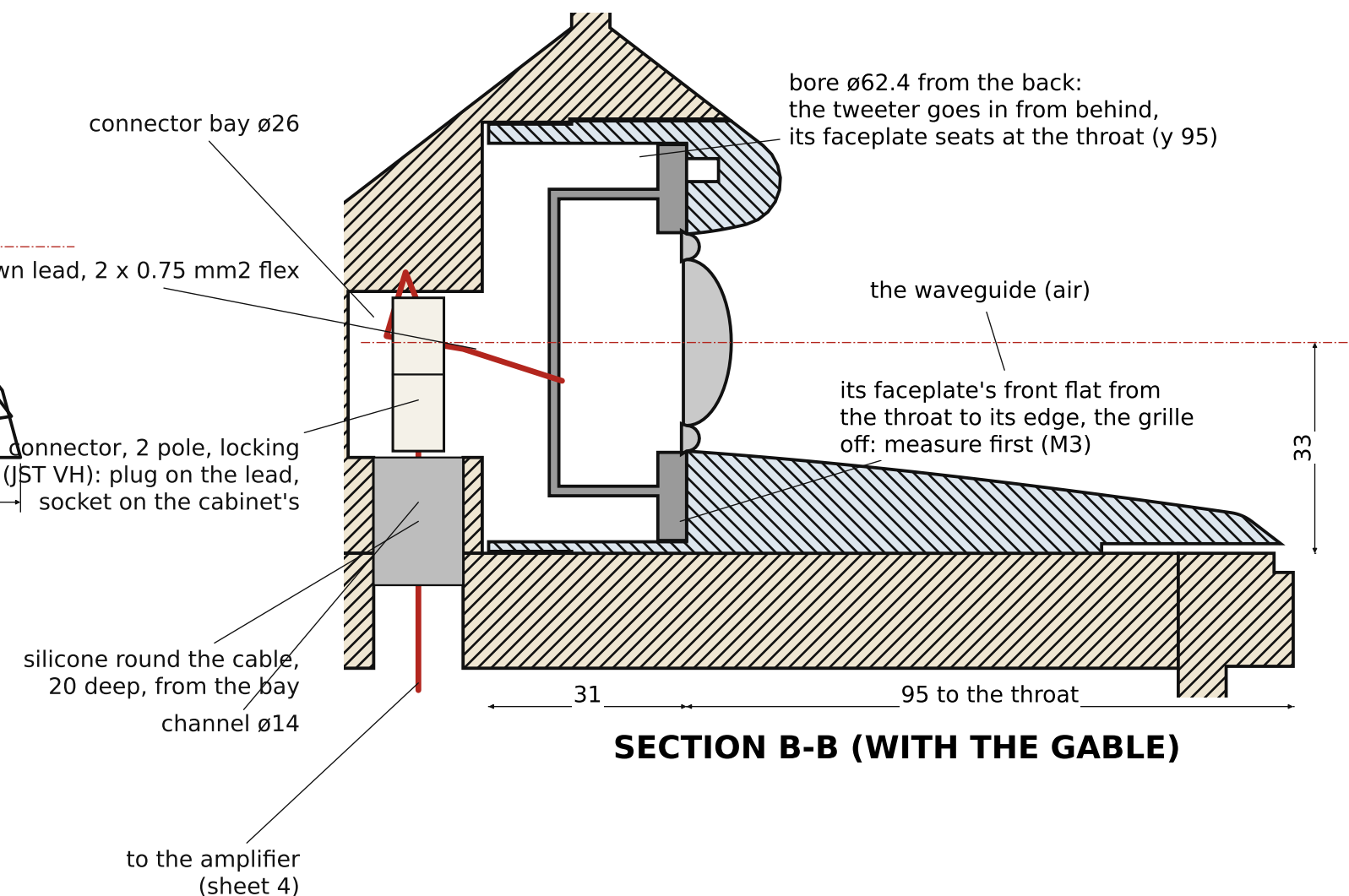
2 of 8
EM-BS-002 rev D



3 x $\varnothing 3.6 \times 5$ holes on $\varnothing 54$ in the seat, one at the top, 120° apart, for M2.5 x 4 knurled brass inserts bonded with epoxy; M2.5 x 8 button heads (ISO 7380, A2) through the faceplate: measure it first



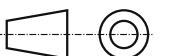
BACK (THE FACE THAT MEETS THE POCKET)



Fitting the tweeter and the insert: notes 3.1 to 3.8, sheet 7

Waveguide insert and tweeter mount

earmilk bookshelf. The insert at 1:1: front, section on the axis, back; the pocket; fitting and service.

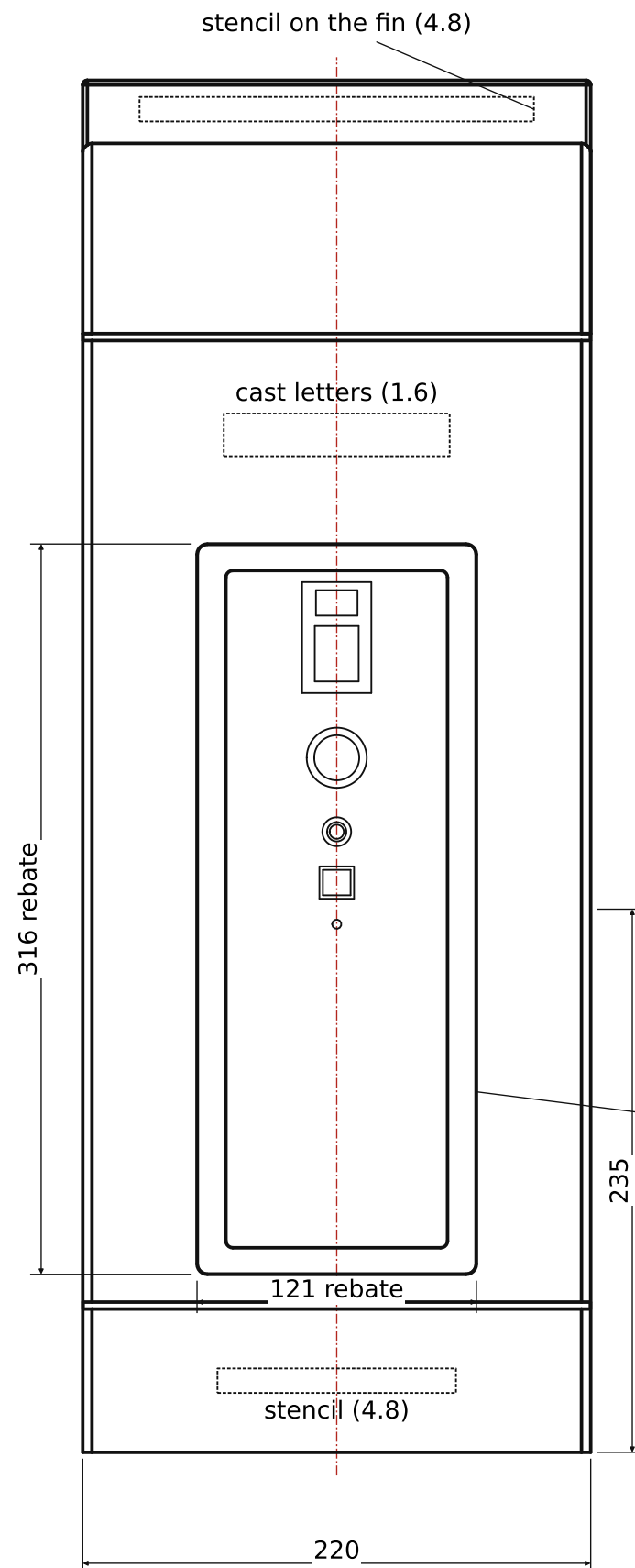


mm | issue 2026-10-09 | from fab/sheets.py (the CAD)

Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

3 of 8

EM-BS-003 rev D



BACK

The Nutrition Facts go on the right side: 150 x 147, top at z 420, centred across its depth, 35 from each vertical corner.

Mains
IEC C14 inlet, switch

Signal in
XLR or RCA
USB for the DSP

Hypex FusionAmp FA122
2 channels, DSP crossover and EQ
125 + 125 W into 4 ohm
upright on the back
in its own sealed box (not airtight itself)
filters: Hypex Filter Design, over USB
mains in on the plate (IEC C14 with switch)

CH1
CH2

Tweeter, Scan-Speak D3004/602200
26 mm dome, faceplate behind the insert
1 mm² pair, red +, black -

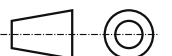
Woofers, SB Acoustics SB17NRX2C35-8
6 in, sealed, DSP shelf to 45 Hz
1.5 mm² pair, red +, black -

Runs, connectors and seals: notes 4.1 to 4.8, sheet 8

plate 120 x 315, in a
121 x 316 x 4.5 rebate;
cut-out 96 x 293, R3 (4.1)

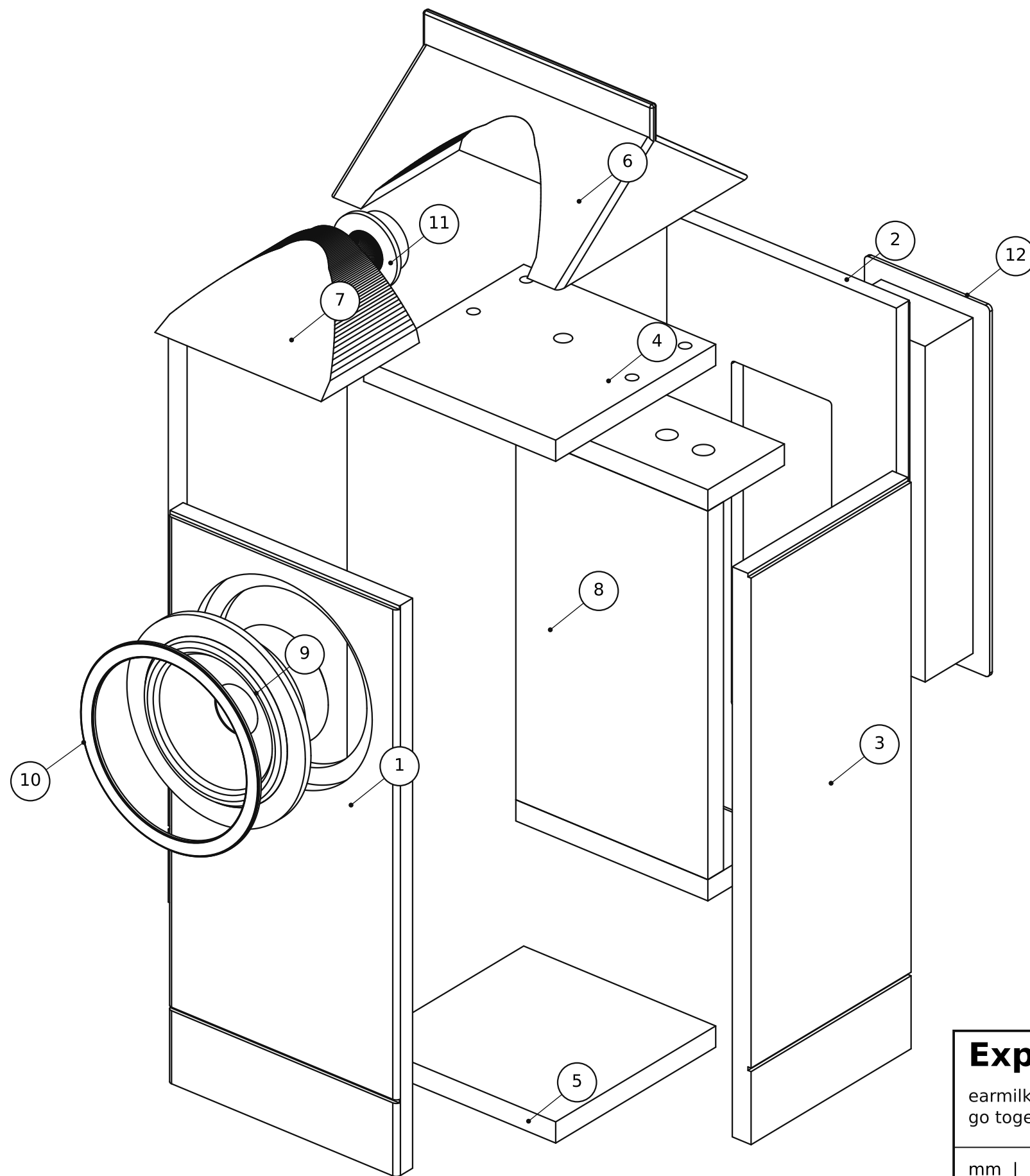
Back and wiring

earmilk bookshelf. The back at 1:3 and the amplifier, drivers, cables and connectors.
Active: the amplifier's DSP is the crossover.



mm | issue 2026-10-09 | from fab/sheets.py (the CAD)
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

4 of 8
EM-BS-004 rev D



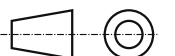
Parts

- ① **Front baffle x1**
18 mm Baltic birch, CNC: the woofer's cut-out, rebate
- ② **Back x1**
18 mm birch: the amplifier's cut-out and rebate
- ③ **Sides x2**
18 mm birch; the Facts printed on the right one
- ④ **Top panel x1**
18 mm birch: the 14 mm cable hole, 4 dowel holes
- ⑤ **Bottom panel x1**
18 mm birch
- ⑥ **Gable block x1**
laminated birch, CNC, or printed in four pieces
- ⑦ **Waveguide insert x1**
SLA tough resin or MJF nylon; magnets, pins
- ⑧ **Amplifier box x3**
birch floor, lid (2 glands) and front; sealed
- ⑨ **Woofer x1**
SB Acoustics SB17NRX2C35-8, 6 in
- ⑩ **Trim ring x1**
printed, satin black, over the frame and screws
- ⑪ **Tweeter x1**
Scan-Speak Illuminator D3004/602200
- ⑫ **Amplifier x1**
Hypex FusionAmp FA122, DSP, 2 channels

Glue-up order: G1 to G5, sheet 8

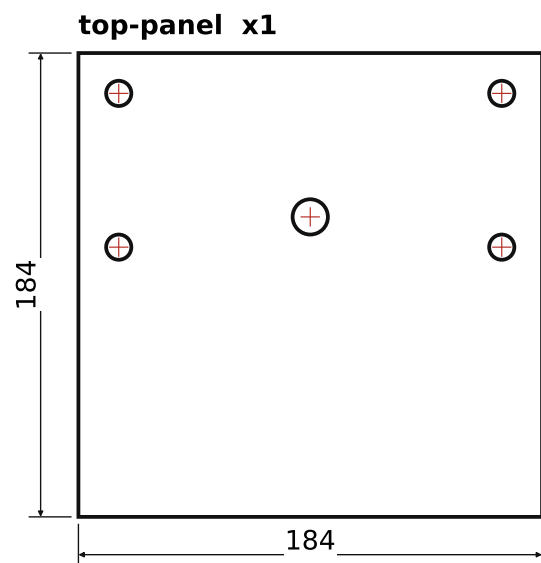
Exploded view and parts

earmilk bookshelf. Isometric, not to scale: the parts pulled apart along the way they go together, and what each one is.

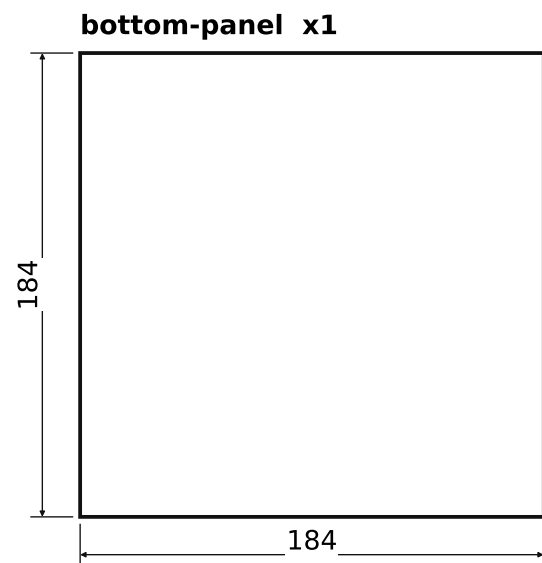


mm | issue 2026-10-09 | from fab/sheets.py (the CAD)
Tolerances: panels ± 0.5 , pockets ± 0.2 , holes $+0.2/0$, prints ± 0.15

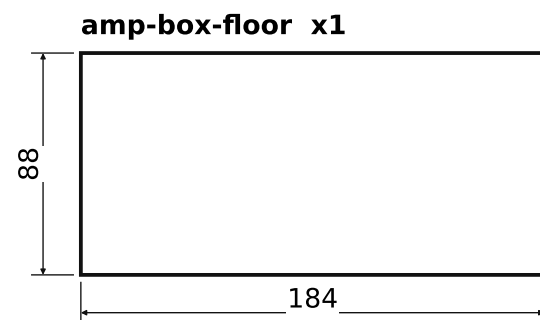
5 of 8
EM-BS-005 rev D



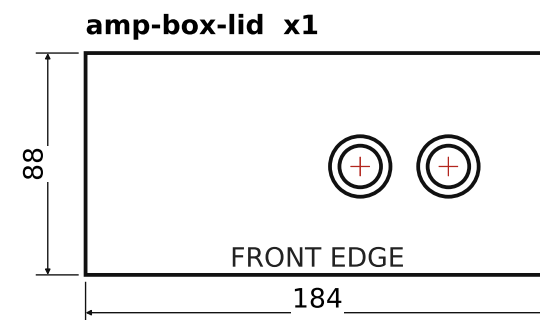
ø14 at (92, 119): cut inside
4 x ø10 at (16, 107), (168, 107), (16, 168),
(168, 168): drill ø10, 10 deep
upper face up (front edge at the bottom of the
drawing); dowels register the gable block; the
tweeter cable's hole is sealed with silicone
from the bay after wiring



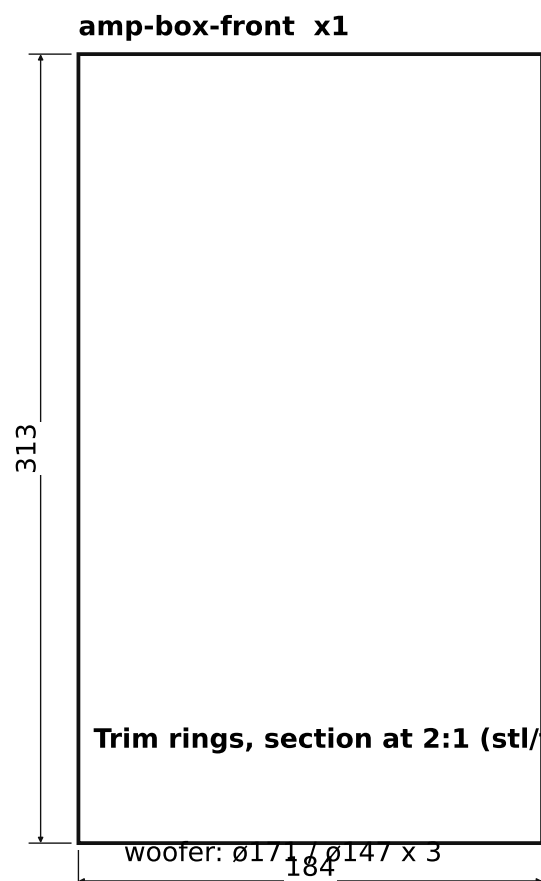
either face



the amplifier box's floor, z 60.5 to 78.5,
against the back



2 x ø16.5 at (144, 43), (109, 43): cut inside
2 x ø24 at (144, 43), (109, 43): pocket 8 mm
deep from the underside, concentric with each
gland's hole (cut the holes, turn the panel
over, centre on them; before the lid goes in)
the amplifier box's lid, z 391.5 to 409.5, its
front edge (the bottom of the drawing) over the
box's front; 2 ø16.5 holes for M16 x 1.5
glands, one per cable, 144, 109 from the left,
43 from the front edge, each nut in a ø24 x 8
counterbore from the underside



the amplifier box's front, between floor and
lid, channel from z150.0 to z168.0, 2 deep
and seal all round

the channel clears the M4 x 20 heads (F1): 8 across, 1.6 high or less

Panel details

earmilk bookshelf. At 1:3: each panel seen from its outer (upper) face, from its
lower-left corner as its DXF; holes from that corner. 18 mm birch.



mm | issue 2026-10-09 | from fab/sheets.py (the CAD)

Tolerances: panels ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

6 of 8

EM-BS-006 rev D

Measure first

- M1** Ply: 18 thick; the sides and inner panels are 220 - 2 x 18 = 184 wide. For a real thickness t: 220 - 2t (heights, pockets and rebates unchanged); set WALL = t in params.py and run fab/build.sh again.
- M2** Woofer (SB17NRX2C35-8): flange 6.5 thick (one source gives 10.9: over 6.5, the rebate would leave under 7.5 for the T-nut barrels, the owner's call: a proud ring or a thicker baffle) sets the rebate, 10.5 deep; the frame's diameter + 1.6 sets the rebate's ø172.6; the cut-out ø144.9; 4 holes on ø159; the surround at its glue line + 2 sets the trim ring's bore, ø147.
- M3** Scan-Speak Illuminator D3004/602200: the faceplate ø62 sets the bore, ø62.4; its hole circle, ø54, and its body, ø48.3 or less, set the seat's screws; its front must be flat from ø34 to ø62 and the grille off (or a seat recess for it); the dome and surround 34 across or less. Print nothing until it is measured.
- M4** Hypex FusionAmp FA122: the plate 120 x 315 sets the rebate, 121 x 316; the module sets the cut-out, 96 x 293; the screw holes come from the plate in hand.

Sheet 1: Notes

- 1.1** Cabinet: 18 mm Baltic birch, glued. The vertical corners and the gable's hips are rounded 4 mm, the fin's edges 2 mm. 3 x 3 mm shadow lines at z 62 and z 481.
- 1.2** The drivers sit flush: each rebate is the measured flange + 1.0 (a 1.5 mm closed-cell foam gasket, pressed) + 3 (the ring): woofer 10.5 for a 6.5 flange; measure each flange before cutting (M2, M3): the rebate is the flange + 4.0 only while 7.5 of birch stays under it for the T-nuts. The screws and T-nuts as F1, pressed in before the glue-up. A printed trim ring over each flange (stl/trim-ring-*.stl, woofer ø171 / ø147 x 3, flat, a channel in its back over the screws' heads: the sections at 2:1 on sheet 6) held by three 5 mm dots of neutral-cure silicone, 0.8 reveal all round. Its inner ø is the surround at its glue line + 2: measure the drivers and print the rings last (the woofer's ring needs a 320 bed, or a print service).
- 1.3** The tweeter is at the throat of a waveguide insert in the roof (sheet 3): axis z 517, 95 behind the front face. The shape is set by simulation (fab/out-bookshelf/acoustics/waveguide).
- 1.4** The back (the amplifier) is drawn on sheet 4 with the wiring; the Nutrition Facts are printed on the right side.
- 1.5** Finish: colour coat and 2K clear; the Facts printed between them.
- 1.6** The cast letters (out/metal/wordmark-letters.*), 1.2 proud: 97.8 x 18.4 of ink, centred across the plinth's front, the badge's centre z 31; on the back the same at z 440. Place and drill them from the 1:1 templates, out/metal/wordmark-template-front.pdf and -back.pdf, after the clear coat.

Sheet 2: The roof

- 2.1** The waveguide insert (light blue) sits in a pocket in the gable block, 0.3 mm clear all round: the pocket +0.2/0, the insert 0/-0.15, so 0.3 to 0.65 a side. Paint only the insert's face: mask its sides, base and back and the pocket's walls. It slides out forward, level, with the tweeter on it.
- 2.2** The tweeter is rear-mounted: it goes in from behind through a ø62.4 bore and its faceplate seats on the ring round the throat, its dome and surround filling the 34 mm throat, screwed from behind through its own holes (sheet 3).
- 2.3** Its lead leaves the boss's open back into the connector bay behind it (ø26 x 22 deep), where it plugs into the cabinet's lead; that runs down a 14 mm channel through the block and the top panel, sealed round the cable with neutral-cure silicone from the bay.
- 2.4** Four magnets and two pins in the insert's back face meet their partners in the pocket's back wall (sheet 3).

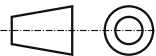
Sheet 3: Fitting the tweeter and the insert

- 3.1** Solder the tweeter's own lead (150 mm of 2 x 0.75 mm2 silicone-insulated flex, twisted (about 2.2 mm each)) to its tabs and crimp the plug half of a two-pole locking connector (JST VH) on its end.
- 3.2** Bond 3 M2.5 x 4 knurled brass inserts into the seat with epoxy (ø3.6 x 5 holes on ø54; a heat-set insert will not melt into cured resin). Pass the tweeter in from behind through the ø62.4 bore, dome first, seat its faceplate on a 0.5 mm foam gasket and screw it to the inserts with M2.5 x 8 button heads (ISO 7380, A2) through its own holes. Measure first: the heads need 0.5 clear of the body (the circle at least the body + 5.7).
- 3.3** Glue the magnets into the insert's back with epoxy, polarity marked, and their partners into the pocket's back wall, opposite poles out. Bond the two pins into the insert with epoxy.
- 3.4** The cabinet's lead runs up the channel and ends in the socket standing in the bay, 63 mm above the top panel's underside (30 past the bay's floor); seal the channel round it with 20 mm of neutral-cure silicone from the bay before the first fitting. The slack is the tweeter's own lead, 150 mm of 2 x 0.75 mm2 silicone-insulated flex, twisted (about 2.2 mm each): hold the insert just in front of its pocket, plug the lead into the socket in the bay, and slide the insert home; the lead folds into the sleeve and the bay.
- 3.5** Slide the insert in, level, until the pins seat and the magnets pull it home: its face flush with the roof, the seam even. Fit: the pocket +0.2/0, the insert 0/-0.15 (0.3 to 0.65 a side); paint only the insert's face.

- 3.6** Service: pull it out by a ribbon loop glued in the 12 x 1.5 x 30 groove under its front edge, unplug, and the tweeter comes out with it.
- 3.7** The pocket (gable-block.step): the insert's outline + 0.3 a side (159.8 wide at its base), from the slope to its back wall at y 113.3; the boss's bore ø69 to y 127; the connector bay ø26 from y 126 to 148 on an axis at z 512; the channel ø14 at x 110, y 137 down to the top panel; internal corners R3 or less (the insert's match); dowel holes ø10 x 20 in the base at (34, 125), (186, 125), (34, 186), (186, 186).
- 3.8** Fixings, from the centreline and the insert's base (mirrored in the pocket's wall): magnets ±41.4 across, 16.9 up; ±41.5 across, 33.0 up; pins ±46.6 across, 27.9 up. Magnets ø8 x 3 in ø8.2 x 3.3 holes, epoxied, poles marked; pins ø4 x 8 bonded with epoxy in ø4.1 x 4.5 in the insert, a slip fit in ø4.2 x 4.5 in the wall (these holes are outside the title block's +0.2/-0).

Notes: measure first; sheets 1 to 3

earmilk bookshelf. What to measure before cutting or printing, and the notes sheets 1 to 3 point to by number.



Sheet 4: Runs, connectors and seals

- 4.1 The amplifier's plate (120 x 315 x 3) lies on 3 mm EPDM tape in its 4.5 deep rebate, R4.5 corners, its module through the 96 x 293 cut-out (R3 corners: a 6 mm cutter or smaller). Drill its screw holes ø3.5 through the 13.5 left under the rebate, from the plate in hand, centred in its flange, 8 or 10 to suit it: 4.3 x 16 self-tapping pan heads (Hypex's 4.3 x 25 kit also works, its tips 7 mm into the sealed box).
- 4.2 Woofer (CH1): from its gland in the amplifier box's lid to the woofer's terminals, 0.4 m of 1.5 mm2.
- 4.3 Tweeter (CH2): from its gland up behind the woofer's magnet to the top panel, through the 14 mm channel in the top panel and the gable block, 0.6 m of 1.0 mm2, ending 63 above the top panel's underside in the socket of a two-pole locking connector (JST VH) standing in the bay behind the insert.
- 4.4 Seal the tweeter cable in its channel with 20 mm of neutral-cure silicone from the bay (sheet 3): the box is sealed, and the insert's pocket is open to the room through its seam.
- 4.5 Round-sheathed cable (H05VV-F 2 x 1.5 and 2 x 1.0 mm2, about 8.0 and 6.4 across), so each gland grips its cable and the silicone seals round the tweeter's; splice any extension of Hypex's harness inside the amplifier's box (crimped butt splices under heat-shrink).
- 4.6 The amplifier box's lid: 2 M16 x 1.5 long-thread (15 mm) cable glands, one per cable (clamping about 4.5 to 10), in ø16.5 holes 144, 109 from the left side's inner face and 43 from the lid's front edge, each lock nut in a ø24 x 8 counterbore under the lid. Fit them after the finish, the bodies from above and the nuts from below through the amplifier's cut-out; feed the cables, then tighten.
- 4.7 The crossover (2.4 kHz, LR4), the woofer's shelf to 45 Hz (+8.3 dB) and the delays are set in the DSP. Wire red to + throughout.
- 4.8 The stencils (out/marks/stencil-*.svg), sprayed through vinyl masks before the clear: SHAKE WELL, 15 mm type, centred on the plinth's back at z 31; OPEN OTHER SIDE, 15 mm type with its arrow, centred on the fin's back face.

Glue-up, in order

- G1 Lay the back face down. Glue both sides and the bottom to it.
- G2 The amplifier box: its floor, front and lid between the sides, against the back; seal its joints.
- G3 Press the T-nuts into the front baffle's inside face, then glue the baffle on.
- G4 The top panel last, inside all four walls.
- G5 Clamp across the sides at every inner panel. Then the gable block on its dowels.

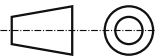
Fixings and the pocket's floor

- F1 Drivers: M4 x 20 low-profile (wafer) button head, A2, the head 1.6 high or less and 8 across or less (ISO 7380 is 2.2 high: the trim ring's channel is 2 deep). M4 T-nuts, flange 12 or less across (woofer), barrel 7 or less (woofer), pressed in before the glue-up.

- F2 Before the gable block goes on, sand the front panel's top edge flush with the top panel to 0.1: the insert sits on both. Mask the pocket's floor with the insert's sides, base and back when you spray: paint only the insert's face.

Notes: sheets 4 to 6; glue-up and fixings

earmilk bookshelf. The notes sheets 4 to 6 point to by number; the order of the glue-up; the fixings.



mm | issue 2026-10-09 | from fab/sheets.py (the CAD)
Tolerances: panels ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15